Class 12 Mathematics

Chapter 7: Integrals

Practice Set A: step-wise solutions

20 questions **54** marks **3** levels

Each row is one line of working. The red number in the right margin is what an examiner gives for that line, as in the CBSE marking scheme (½ is half a mark). Write every line that carries a mark. ✓ ½

Level 1

8 marks

1 **MCQ** PYQ-style — 1 mark

$\int e^x \sec x\,(1+\tan x)\,dx$ is equal to:

(A) $e^x \cos x + C$

(B) $e^x \sec x + C$

(C) $e^x \sin x + C$

(D) $e^x \tan x + C$

Formula / idea: $\int e^x\,[f(x)+f'(x)]\,dx = e^x f(x) + C$

1 Open the bracket
$e^x \sec x\,(1+\tan x) = e^x\,(\sec x + \sec x \tan x)$

2 Take $f(x) = \sec x$
$f'(x) = \sec x \tan x$

3 So the integral is
$\int e^x(\sec x + \sec x \tan x)\,dx = e^x \sec x + C$ ✓ 1

Answer **(B)** $e^x \sec x + C$

Watch out: Picking (D): $f(x) = \tan x$ needs $f'(x) = \sec^2 x$, which is not in the bracket.

2 MCQ CBSE SQP 2025-26 1 mark

If $\int x^3 \sin^4(x^4)\cos(x^4)\,dx = a\sin^5(x^4) + C$, then a is equal to:

(A) $-\frac{1}{10}$

(B) $\frac{1}{20}$

(C) $\frac{1}{4}$

(D) $\frac{1}{5}$

Formula / idea: Put the inside function $= t$; its derivative must be present (up to a constant).

1 Put $\sin(x^4) = t$

$\cos(x^4)\cdot 4x^3\,dx = dt \;\Rightarrow\; x^3\cos(x^4)\,dx = \frac{dt}{4}$

2 Substitute

$\int t^4 \cdot \frac{dt}{4} = \frac{1}{4}\cdot\frac{t^5}{5} + C = \frac{t^5}{20} + C$

3 Put back $t = \sin(x^4)$ and compare ✓ 1

$\frac{1}{20}\sin^5(x^4) + C \;\Rightarrow\; a = \frac{1}{20}$

Answer **(B)** $\frac{1}{20}$

Watch out: Forgetting the $\frac{1}{4}$ from $d(x^4) = 4x^3\,dx$ and choosing (D) $\frac{1}{5}$.

3 MCQ PYQ-style 1 mark

The value of $\int_0^{\pi/2} x\sin x\,dx$ is:

(A) 0

(B) 1

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{2} - 1$

Formula / idea: By parts: $\int u\,v\,dx = u\int v\,dx - \int\left(u'\int v\,dx\right)dx$

1 Integrate by parts, taking $u = x$, $v = \sin x$

$\int x\sin x\,dx = -x\cos x + \int \cos x\,dx = -x\cos x + \sin x$

2 Put the limits

$$\left[-x\cos x+\sin x\right]_0^{\pi/2}=\left(-\frac{\pi}{2}\cdot 0+1\right)-(0+0)$$

3 Simplify

$$=1$$

✓ 1

Answer **(B)** 1

Watch out: Writing $\int \sin x\,dx=\cos x$; the correct result is $-\cos x$.

4 **MCQ** CBSE SQP 2024-25 1 mark

$\int \dfrac{dx}{x^3\left(1+x^4\right)^{1/2}}$ equals:

(A) $-\dfrac{1}{2x^2}\sqrt{1+x^4}+C$

(B) $\dfrac{1}{2x}\sqrt{1+x^4}+C$

(C) $-\dfrac{1}{4x}\sqrt{1+x^4}+C$

(D) $\dfrac{1}{4x^2}\sqrt{1+x^4}+C$

Formula / idea: Take the highest power of x out of the root, then substitute.

1 Take x^4 out of the root; $\sqrt{x^4}=x^2$

$$\int\frac{dx}{x^3\cdot x^2\sqrt{1+x^{-4}}}=\int\frac{x^{-5}\,dx}{\sqrt{1+x^{-4}}}$$

2 Put $1+x^{-4}=t$

$$-4x^{-5}\,dx=dt$$

3 Substitute

$$-\frac{1}{4}\int t^{-1/2}\,dt=-\frac{1}{2}\sqrt{t}+C$$

4 Put back $t=\dfrac{1+x^4}{x^4}$

$$-\frac{1}{2}\cdot\frac{\sqrt{1+x^4}}{x^2}+C=-\frac{1}{2x^2}\sqrt{1+x^4}+C$$

✓ 1

Answer **(A)** $-\dfrac{1}{2x^2}\sqrt{1+x^4}+C$

Watch out: Dropping the minus sign in $dt=-4x^{-5}\,dx$, or the $\frac{1}{4}$ with it; the answer is $-\dfrac{1}{2x^2}\sqrt{1+x^4}$.

5 MCQ CBSE SQP 2025-26 1 mark

If $f(a+b-x)=f(x)$, then $\int_a^b x\,f(x)\,dx$ is equal to:

(A) $\dfrac{a+b}{2}\int_a^b f(b-x)\,dx$

(B) $\dfrac{a+b}{2}\int_a^b f(a-x)\,dx$

(C) $\dfrac{b-a}{2}\int_a^b f(x)\,dx$

(D) $\dfrac{a+b}{2}\int_a^b f(x)\,dx$

Formula / idea: $\int_a^b f(x)\,dx=\int_a^b f(a+b-x)\,dx$

1 Let

$$I=\int_a^b x\,f(x)\,dx \quad \ldots(1)$$

2 Replace x by $a+b-x$ and use $f(a+b-x)=f(x)$

$$I=\int_a^b (a+b-x)\,f(x)\,dx \quad \ldots(2)$$

3 Add (1) and (2) ✓ 1

$$2I=(a+b)\int_a^b f(x)\,dx \;\Rightarrow\; I=\frac{a+b}{2}\int_a^b f(x)\,dx$$

Answer **(D)** $\dfrac{a+b}{2}\int_a^b f(x)\,dx$

Watch out: Choosing (C): the factor is $a+b$ (from adding x and $a+b-x$), not $b-a$.

6 **Assertion-Reason** Board-pattern 1 mark

Assertion (A): $\int_{-1}^{1} \log\left(\frac{2-x}{2+x}\right) dx = 0$. **Reason (R):** If f is an even function, then $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) **Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).**

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Formula / idea: If $f(-x) = -f(x)$, then $\int_{-a}^{a} f(x)\,dx = 0$

1	Check A: let $f(x) = \log\frac{2-x}{2+x}$ and replace x by $-x$ $f(-x) = \log\frac{2+x}{2-x} = -\log\frac{2-x}{2+x} = -f(x)$	
2	So f is odd; its integral over $[-1, 1]$ is 0. A is true.	
3	R is the standard property of even functions. R is true.	
4	A uses the odd-function property, not R. So R does not explain A.	
5	Hence the correct option is (B).	✓ 1

Answer **(B) Both true; R is not the correct explanation of A**

Watch out: Marking (A) because both statements are true, without checking that R is about even functions.

7 **Short answer** PYQ-style 2 marks

Find $\int \frac{\sin x - \cos x}{\sqrt{1+\sin 2x}}\,dx,\ 0 < x < \frac{\pi}{2}$.

Formula / idea: $1 + \sin 2x = (\sin x + \cos x)^2$

1	Write $1 = \sin^2 x + \cos^2 x$ and $\sin 2x = 2\sin x \cos x$ $1 + \sin 2x = (\sin x + \cos x)^2$	✓ ½
2	For $0 < x < \frac{\pi}{2}$, $\sin x + \cos x > 0$ $I = \int \frac{\sin x - \cos x}{\sin x + \cos x}\,dx$	✓ ½
3	Put $\sin x + \cos x = t$ $(\cos x - \sin x)\,dx = dt \ \Rightarrow\ (\sin x - \cos x)\,dx = -dt$	✓ ½

4 Integrate and put back t ✓ ½

$$I = -\int \frac{dt}{t} = -\log|t| + C = -\log|\sin x + \cos x| + C$$

Answer $-\log|\sin x + \cos x| + C$

Watch out: Losing the minus sign: the derivative of $\sin x + \cos x$ is $\cos x - \sin x$, the negative of the numerator.

Level 2

19 marks

8 **Short answer** PYQ-style 2 marks

Evaluate: $\int_0^{\pi/4} \tan^2 x \, dx.$

Formula / idea: $\tan^2 x = \sec^2 x - 1$

1 Use $\tan^2 x = \sec^2 x - 1$ ✓ ½

$$I = \int_0^{\pi/4} (\sec^2 x - 1)\, dx$$

2 Integrate ✓ ½

$$I = \left[\tan x - x\right]_0^{\pi/4}$$

3 Put the limits ✓ 1

$$I = \left(\tan\frac{\pi}{4} - \frac{\pi}{4}\right) - (0 - 0) = 1 - \frac{\pi}{4}$$

Answer $1 - \frac{\pi}{4}$

Watch out: Writing $\int \tan^2 x \, dx = \frac{\tan^3 x}{3}$; there is no $\sec^2 x$ factor for that.

9 **Short answer** CBSE SQP 2023-24 2 marks

Find: $\int \sqrt{\frac{x}{1-x^3}}\, dx;\ x \in (0,1).$

Formula / idea: $\int \frac{dt}{\sqrt{1-t^2}} = \sin^{-1} t + C$

1 Write $x^3 = \left(x^{3/2}\right)^2$ ✓ ½

$$I = \int \frac{\sqrt{x}}{\sqrt{1-\left(x^{3/2}\right)^2}}\, dx$$

2 Put $x^{3/2} = t$
$\frac{3}{2}x^{1/2}\,dx = dt \;\Rightarrow\; \sqrt{x}\,dx = \frac{2}{3}\,dt$ ✓ ½

3 Substitute
$I = \frac{2}{3}\int \frac{dt}{\sqrt{1-t^2}} = \frac{2}{3}\sin^{-1} t + C$ ✓ ½

4 Put back $t = x^{3/2}$
$I = \frac{2}{3}\sin^{-1}\left(x^{3/2}\right) + C$ ✓ ½

Answer $\frac{2}{3}\sin^{-1}\left(x^{3/2}\right) + C$

Watch out: Putting $x^3 = t$; then $\sqrt{x}$ in the numerator does not match $3x^2\,dx$.

10 **Short answer** CBSE SQP 2023-24 3 marks

Find: $\int \frac{2x^2+3}{x^2\left(x^2+9\right)}\,dx;\; x \neq 0.$

Formula / idea: Only x^2 appears, so put $x^2 = y$ just to find the partial fractions.

1 Put $x^2 = y$ (for partial fractions only)
$\frac{2y+3}{y(y+9)} = \frac{A}{y} + \frac{B}{y+9}$ ✓ ½

2 Multiply by $y(y+9)$
$2y + 3 = A(y+9) + By$ ✓ ½

3 Put $y = 0$, then $y = -9$
$A = \frac{3}{9} = \frac{1}{3}, \qquad B = \frac{-15}{-9} = \frac{5}{3}$ ✓ 1

4 Put back $y = x^2$ and integrate
$I = \frac{1}{3}\int x^{-2}\,dx + \frac{5}{3}\int \frac{dx}{x^2+3^2}$ ✓ ½

5 Use $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a}$
$I = -\frac{1}{3x} + \frac{5}{9}\tan^{-1}\frac{x}{3} + C$ ✓ ½

Answer $-\frac{1}{3x} + \frac{5}{9}\tan^{-1}\frac{x}{3} + C$

Watch out: Integrating $\frac{1}{x^2+9}$ as $\tan^{-1}\frac{x}{3}$ and dropping the factor $\frac{1}{3}$.

11 **Short answer** PYQ-style 3 marks

Evaluate: $\displaystyle\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$.

Formula / idea: $\displaystyle\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$; here $a+b=\frac{\pi}{2}$.

1 Write $\tan x = \dfrac{\sin x}{\cos x}$

$$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx \quad \ldots(1)$$

✓ ½

2 Replace x by $\dfrac{\pi}{6}+\dfrac{\pi}{3}-x = \dfrac{\pi}{2}-x$

$$I = \int_{\pi/6}^{\pi/3} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}\,dx \quad \ldots(2)$$

✓ 1

3 Add (1) and (2)

$$2I = \int_{\pi/6}^{\pi/3} 1\,dx$$

✓ ½

4 Integrate

$$2I = \frac{\pi}{3}-\frac{\pi}{6} = \frac{\pi}{6}$$

✓ ½

5 Hence

$$I = \frac{\pi}{12}$$

✓ ½

Answer $\dfrac{\pi}{12}$

Watch out: Replacing x by $\dfrac{\pi}{3}-x$; with limits a to b it must be $a+b-x$.

12 **Short answer** CBSE SQP 2024-25 3 marks

Evaluate: $\displaystyle\int \left\{\frac{1}{\log x}-\frac{1}{(\log x)^2}\right\} dx;\ x>1.$

Formula / idea: $\displaystyle\int e^t\,[f(t)+f'(t)]\,dt = e^t f(t)+C$

1 Put $\log x = t$

$$x = e^t, \qquad dx = e^t\,dt$$

✓ 1

2 Substitute

$$I = \int e^t\left(\frac{1}{t}-\frac{1}{t^2}\right)dt$$

✓ ½

3 Take $f(t) = \dfrac{1}{t}$

$f'(t) = -\dfrac{1}{t^2}$ ✓ ½

4 Use the $e^t[f + f']$ result

$I = e^t \cdot \dfrac{1}{t} + C$ ✓ ½

5 Put back $e^t = x$, $t = \log x$

$I = \dfrac{x}{\log x} + C$ ✓ ½

Answer $\dfrac{x}{\log x} + C$

Watch out: Forgetting $dx = e^t\,dt$, which is what creates the e^t factor.

13 Short answer CBSE SQP 2025-26 — 3 marks

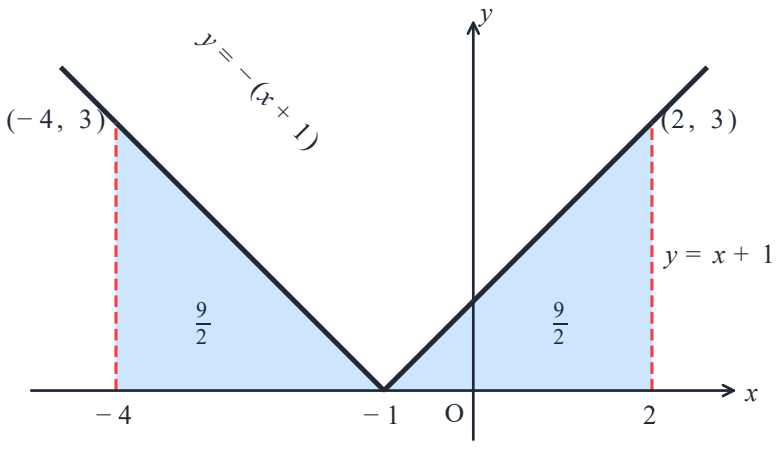
Sketch the graph of $y = |x + 1|$. Evaluate $\displaystyle\int_{-4}^{2} |x + 1|\,dx$. What does the value of this integral represent on the graph?

Formula / idea: Split a modulus integral at the point where the expression inside is 0.

1 Remove the modulus ($x + 1 = 0$ at $x = -1$) ✓ ½

$$|x+1| = \begin{cases} -(x+1), & x < -1 \\ x+1, & x \geq -1 \end{cases}$$

2 Sketch: a V-shape with vertex at $(-1, 0)$ ✓ ½



3 Split the integral at $x = -1$ ✓ ½

$$I = \int_{-4}^{-1} -(x+1)\,dx + \int_{-1}^{2} (x+1)\,dx$$

4 First part

$$-\left[\frac{(x+1)^2}{2}\right]_{-4}^{-1} = -\left(0-\frac{9}{2}\right) = \frac{9}{2}$$

✓ ½

5 Second part

$$\left[\frac{(x+1)^2}{2}\right]_{-1}^{2} = \frac{9}{2} - 0 = \frac{9}{2}$$

✓ ½

6 Add; the value is the shaded area in the graph

$$I = \frac{9}{2} + \frac{9}{2} = 9 \text{ sq. units}$$

✓ ½

Answer 9**; it is the area bounded by** $y = |x+1|$**, the** x**-axis and the lines** $x = -4$, $x = 2$**.**

Watch out: Integrating $(x+1)$ straight from -4 to 2, which gives 3 because the left part counts as negative.

14 **Short answer** CBSE 2025 3 marks

Evaluate: $\displaystyle\int_0^{\pi/4} \frac{dx}{\cos^3 x\sqrt{2\sin 2x}}$.

Formula / idea: Change everything to $\tan x$ and $\sec^2 x$, then put $\tan x = t$.

1 Use $\sin 2x = 2\sin x\cos x = 2\tan x\cos^2 x$

$$\sqrt{2\sin 2x} = \sqrt{4\tan x\cos^2 x} = 2\cos x\sqrt{\tan x}$$

✓ ½

2 So the integrand becomes

$$\frac{1}{2\cos^4 x\sqrt{\tan x}} = \frac{\sec^2 x\,(1+\tan^2 x)}{2\sqrt{\tan x}}$$

✓ ½

3 Put $\tan x = t$, $\sec^2 x\,dx = dt$

$$x = 0 \Rightarrow t = 0, \qquad x = \frac{\pi}{4} \Rightarrow t = 1$$

✓ ½

4 Change the integral

$$I = \frac{1}{2}\int_0^1 \left(t^{-1/2} + t^{3/2}\right) dt$$

✓ ½

5 Integrate

$$I = \frac{1}{2}\left[2t^{1/2} + \frac{2}{5}t^{5/2}\right]_0^1$$

✓ ½

6 Put the limits

$$I = \frac{1}{2}\left(2 + \frac{2}{5}\right) = \frac{6}{5}$$

✓ ½

Answer $\dfrac{6}{5}$

Watch out: Writing $\sec^4 x = \sec^2 x$; one $\sec^2 x$ must become $1+\tan^2 x$.

Level 3

27 marks

15 **Long answer** PYQ-style 5 marks

Evaluate: $\displaystyle\int_0^{\pi} \frac{x\tan x}{\sec x+\tan x}\,dx.$

Formula / idea: $\displaystyle\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ removes the factor x.

1	Let $I = \displaystyle\int_0^{\pi} \frac{x\tan x}{\sec x+\tan x}\,dx \quad \ldots(1)$	
2	Replace x by $\pi - x$; tan and sec both change sign $I = \displaystyle\int_0^{\pi} \frac{(\pi-x)(-\tan x)}{-\sec x-\tan x}\,dx = \int_0^{\pi} \frac{(\pi-x)\tan x}{\sec x+\tan x}\,dx \quad \ldots(2)$	✓ 1
3	Add (1) and (2) $2I = \pi\displaystyle\int_0^{\pi} \frac{\tan x}{\sec x+\tan x}\,dx$	✓ ½
4	Multiply top and bottom by $\cos x$ $\dfrac{\tan x}{\sec x+\tan x} = \dfrac{\sin x}{1+\sin x} = 1 - \dfrac{1}{1+\sin x}$	✓ ½
5	Multiply by $\dfrac{1-\sin x}{1-\sin x}$ $\dfrac{1}{1+\sin x} = \dfrac{1-\sin x}{\cos^2 x} = \sec^2 x - \sec x\tan x$	✓ ½
6	So $2I = \pi\displaystyle\int_0^{\pi} \left(1 - \sec^2 x + \sec x\tan x\right)dx$	✓ ½
7	Integrate $2I = \pi\left[x - \tan x + \sec x\right]_0^{\pi}$ $\sec x - \tan x = \dfrac{\cos x}{1+\sin x}$ has no break at $\dfrac{\pi}{2}$	✓ 1
8	Put the limits ($\sec\pi = -1$, $\sec 0 = 1$) $2I = \pi\left[(\pi - 0 - 1) - (0 - 0 + 1)\right] = \pi(\pi-2)$	✓ ½
9	Hence $I = \dfrac{\pi(\pi-2)}{2}$	✓ ½

Answer $\dfrac{\pi(\pi-2)}{2}$

Watch out: Taking $\sec(\pi - x) = \sec x$; it is $-\sec x$, like $\cos(\pi - x) = -\cos x$.

16 **Long answer** CBSE SQP 2025-26 5 marks

Find: $\int \frac{(3\sin\theta - 2)\cos\theta}{5 - \cos^2\theta - 4\sin\theta}\, d\theta.$

Formula / idea: Repeated factor $(t-a)^2$ needs two fractions: $\frac{A}{t-a} + \frac{B}{(t-a)^2}$.

1	Put $\sin\theta = t$ $\cos\theta\, d\theta = dt$	✓ 1
2	Use $\cos^2\theta = 1 - t^2$ in the denominator $5 - (1 - t^2) - 4t = t^2 - 4t + 4 = (t-2)^2$	✓ ½
3	So $I = \int \frac{3t - 2}{(t-2)^2}\, dt$	✓ ½
4	Write as partial fractions $\frac{3t-2}{(t-2)^2} = \frac{A}{t-2} + \frac{B}{(t-2)^2} \Rightarrow 3t - 2 = A(t-2) + B$	✓ ½
5	Compare coefficient of t; then put $t = 2$ $A = 3, \qquad B = 4$	✓ 1
6	Integrate each fraction $I = 3\int \frac{dt}{t-2} + 4\int (t-2)^{-2}\, dt = 3\log\lvert t-2\rvert - \frac{4}{t-2} + C$	✓ 1
7	Put back $t = \sin\theta$; note $\lvert\sin\theta - 2\rvert = 2 - \sin\theta$ $I = 3\log(2 - \sin\theta) + \frac{4}{2 - \sin\theta} + C$	✓ ½

Answer $3\log(2 - \sin\theta) + \frac{4}{2 - \sin\theta} + C$

Watch out: Writing $\int \frac{dt}{(t-2)^2} = \log(t-2)^2$; it is $-\frac{1}{t-2}$.

17 **Long answer** CBSE SQP 2025-26 5 marks

Evaluate: $\int_0^1 \frac{\log(1+x)}{1+x^2}\, dx.$

Formula / idea: Put $x = \tan\theta$, then use $\int_0^a f(x)\, dx = \int_0^a f(a-x)\, dx$.

1	Put $x = \tan\theta$, $dx = \sec^2\theta\, d\theta$ $x = 0 \Rightarrow \theta = 0, \qquad x = 1 \Rightarrow \theta = \frac{\pi}{4}$	✓ 1

2 Use $1+\tan^2\theta=\sec^2\theta$

$$I=\int_0^{\pi/4}\frac{\log(1+\tan\theta)}{\sec^2\theta}\sec^2\theta\,d\theta=\int_0^{\pi/4}\log(1+\tan\theta)\,d\theta \quad \dots(1)$$

✓ ½

3 Replace θ by $\frac{\pi}{4}-\theta$

$$1+\tan\left(\frac{\pi}{4}-\theta\right)=1+\frac{1-\tan\theta}{1+\tan\theta}=\frac{2}{1+\tan\theta}$$

✓ 1

4 So

$$I=\int_0^{\pi/4}\log\frac{2}{1+\tan\theta}\,d\theta=\int_0^{\pi/4}[\log 2-\log(1+\tan\theta)]\,d\theta$$

✓ ½

5 Use (1) for the second integral

$$I=\frac{\pi}{4}\log 2-I$$

✓ 1

6 Hence

$$2I=\frac{\pi}{4}\log 2\ \Rightarrow\ I=\frac{\pi}{8}\log 2$$

✓ 1

Answer $\frac{\pi}{8}\log 2$

Watch out: Not changing the limits: after $x=\tan\theta$ the upper limit is $\frac{\pi}{4}$, not 1.

18 Short answer PYQ-style — 3 marks

Find: $\int\frac{x+3}{\sqrt{5-4x-x^2}}\,dx.$

Formula / idea: Numerator $=A\cdot\frac{d}{dx}(\text{quadratic})+B$; then $\int\frac{dx}{\sqrt{a^2-x^2}}=\sin^{-1}\frac{x}{a}$

1 Write $x+3=A\frac{d}{dx}(5-4x-x^2)+B$

$$x+3=A(-4-2x)+B$$

2 Compare coefficients: $-2A=1,\ -4A+B=3$

$$A=-\frac{1}{2},\qquad B=1$$

✓ ½

3 Split the integral

$$I=-\frac{1}{2}\int\frac{-4-2x}{\sqrt{5-4x-x^2}}\,dx+\int\frac{dx}{\sqrt{5-4x-x^2}}=-\frac{1}{2}I_1+I_2$$

✓ ½

4 In I_1 put $5-4x-x^2=t$, so $(-4-2x)\,dx=dt$

$$I_1=\int\frac{dt}{\sqrt{t}}=2\sqrt{t}=2\sqrt{5-4x-x^2}$$

✓ ½

5 In I_2 complete the square

$$5-4x-x^2=9-(x+2)^2=3^2-(x+2)^2$$

✓ ½

6	Use $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a}$ $I_2 = \sin^{-1}\left(\frac{x+2}{3}\right)$	✓ ½
7	Hence $I = -\sqrt{5-4x-x^2} + \sin^{-1}\left(\frac{x+2}{3}\right) + C$	✓ ½

Answer $-\sqrt{5-4x-x^2} + \sin^{-1}\left(\frac{x+2}{3}\right) + C$

Watch out: Completing the square wrongly as $9+(x+2)^2$; take out the minus sign first: $-(x^2+4x-5)$.

19 **Long answer** PYQ-style 5 marks

Evaluate: $\int_0^{\pi/4} \frac{\sin x + \cos x}{9 + 16\sin 2x}\,dx.$

Formula / idea: Put $\sin x - \cos x = t$; then $\sin 2x = 1 - t^2$.

1	Put $\sin x - \cos x = t$ $(\cos x + \sin x)\,dx = dt$	✓ 1
2	Square $\sin x - \cos x = t$ $1 - \sin 2x = t^2 \;\Rightarrow\; \sin 2x = 1 - t^2$	✓ 1
3	Change the limits $x = 0 \Rightarrow t = -1, \qquad x = \frac{\pi}{4} \Rightarrow t = 0$	✓ ½
4	Simplify the denominator $9 + 16(1-t^2) = 25 - 16t^2$	✓ ½
5	So $I = \int_{-1}^{0} \frac{dt}{25-16t^2} = \frac{1}{16}\int_{-1}^{0} \frac{dt}{\left(\frac{5}{4}\right)^2 - t^2}$	✓ ½
6	Use $\int \frac{dt}{a^2-t^2} = \frac{1}{2a}\log\left\lvert\frac{a+t}{a-t}\right\rvert$ with $a = \frac{5}{4}$ $I = \frac{1}{16}\cdot\frac{2}{5}\left[\log\left\lvert\frac{5+4t}{5-4t}\right\rvert\right]_{-1}^{0} = \frac{1}{40}\left[\log\left\lvert\frac{5+4t}{5-4t}\right\rvert\right]_{-1}^{0}$	✓ 1
7	Put the limits $I = \frac{1}{40}\left[\log 1 - \log\frac{1}{9}\right] = \frac{1}{40}\log 9$	✓ ½

Answer $\frac{1}{40}\log 9$

Watch out: Putting $\sin x + \cos x = t$; its derivative is $\cos x - \sin x$, which does not match the numerator.

20 **Case study** Board-pattern 4 marks

Case Study: While revising definite integrals, a student notes the property $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$. When the lower limit is 0, it becomes $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$. Use these properties to answer the following.

(i) Evaluate $\displaystyle\int_2^8 \frac{\sqrt{10-x}}{\sqrt{x}+\sqrt{10-x}}\,dx$. 1 mark

(ii) Evaluate $\displaystyle\int_0^{\pi/2} \log(\tan x)\,dx$. 1 mark

(iii) Evaluate $\displaystyle\int_0^{\pi/2} (2\log\sin x - \log\sin 2x)\,dx$. 2 marks

Formula / idea: $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$; on $\left[0, \frac{\pi}{2}\right]$, sin and cos swap.

(i) 1 mark

1 Let

$$I = \int_2^8 \frac{\sqrt{10-x}}{\sqrt{x}+\sqrt{10-x}}\,dx \quad \ldots(1)$$

2 Replace x by $2+8-x = 10-x$ ✓ ½

$$I = \int_2^8 \frac{\sqrt{x}}{\sqrt{10-x}+\sqrt{x}}\,dx \quad \ldots(2)$$

3 Add (1) and (2) ✓ ½

$$2I = \int_2^8 1\,dx = 6 \;\Rightarrow\; I = 3$$

Answer (i) 3

(ii) 1 mark

1 Let

$$I = \int_0^{\pi/2} \log(\tan x)\,dx \quad \ldots(1)$$

2 Replace x by $\frac{\pi}{2} - x$ ✓ ½

$$I = \int_0^{\pi/2} \log(\cot x)\,dx = -\int_0^{\pi/2} \log(\tan x)\,dx = -I$$

3 So ✓ ½

$$2I = 0 \;\Rightarrow\; I = 0$$

Answer (ii) 0

(iii) 2 marks

1	Use $\sin 2x = 2\sin x \cos x$ $2\log\sin x - \log\sin 2x = \log\sin x - \log\cos x - \log 2$	✓ ½
2	So $I = \int_0^{\pi/2} \log\sin x\, dx - \int_0^{\pi/2} \log\cos x\, dx - \frac{\pi}{2}\log 2$	✓ ½
3	Replace x by $\frac{\pi}{2} - x$ in the first integral $\int_0^{\pi/2} \log\sin x\, dx = \int_0^{\pi/2} \log\cos x\, dx$	✓ ½
4	The first two integrals cancel $I = -\frac{\pi}{2}\log 2$	✓ ½

Answer (iii) $-\frac{\pi}{2}\log 2$

Watch out: In (iii), trying to find $\int_0^{\pi/2} \log\sin x\, dx$ separately; it is not needed, it cancels.